

ALDA: An Adaptive Layout Design Assistant for Diverse Posters Throughout the Design Process

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ABSTRACT

Layout generation is important in the field of graphic design and has attracted intensive research attention recently. To further prompt human-computer interactions, we construct the ALDA to assist users throughout the design process, which achieves adaptive and diverse content expansions upon only one element for beginners and generates high-quality posters subsequently. Specifically, to obtain diverse contents, we propose aesthetic-aware design graphs (AGs) for effective poster representation and propose a self-constrained blending strategy upon related examples. In addition, we build a novel layout generator to better arrange elements conditioned on our AGs. Finally, we implement ALDA as an online tool with a set of controllable factors to enhance its practicality.

CCS CONCEPTS

• **Applied computing** → **Performing arts**; • **Human-centered computing** → **Interface design prototyping**.

KEYWORDS

graph expansion, human-computer interactive design, aesthetic representation, layout generation, transformer

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1 INTRODUCTION

Layout generation is important in the field of graphic design. A trending way to generate reasonable and complex layouts is taking scene graphs (SGs) as input [1, 3, 10] to provide more detailed structural information, and the content expansion task aims to reduce users' burden by automatically recovering incomplete input

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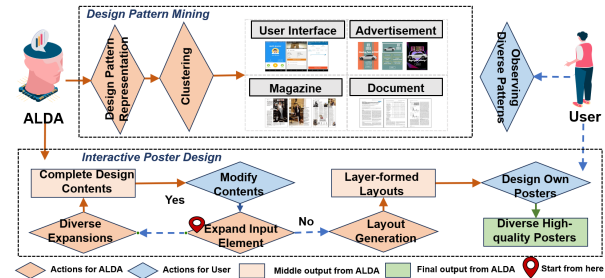


Figure 1: System Framework of ALDA.

graphs. Recent studies depend on deep-learning methods to predict unknown content conditioned on the rest of the input graph [3, 10, 11], which can only deal with a small amount of missing content (e.g. one or two nodes or edges) for accurate prediction.

However, designers usually start work from a single element (e.g. the main product for advertisements), and collect real-world examples to draw inspiration for diverse variations [2, 9]. Therefore, this paper introduces ALDA, an Adaptive Layout Design Assistant, to assist designers from the beginning to the last diverse complete posters. Our ALDA is equipped with the following capabilities.

- Mines classic design patterns from real-world examples to provide users with comprehensive inspiration.
- Handles both limited (even single) and complex (rich elements and constraints) input for diverse content expansions.
- Generates reasonable high-quality layer-formed layouts conditioned on the expanded contents.

2 DEMONSTRATION

Fig. 1 introduces the system framework of our ALDA, which is equipped with two key capabilities including Design Pattern Mining (DPM) and Interactive Poster Design (IPD). Specifically, we construct ALDA with an online tool as shown in Fig. 2. The first two interfaces are introduced for DPM by displaying image clustering results and their details, which aims to provide beginners with specific choices when they do not come up with clear craft. The third page accomplishes interactive content expansions, where users provide one element, select preference contents provided by ALDA, and obtain final satisfied results by requiring our system to re-expand the current content several times. The last page illustrates the generated layer-formed layouts and other design tools for a further touch-up. Please check our video for more details.



Figure 2: Our Human-Computer Interactive Design Tool.

3 METHOD

This section introduces the core techniques in ALDA and details are provided in the supplementary file.

3.1 Design Pattern Mining

The effective representation of design information for layouts is crucial for pattern mining. Inspired by the artistic theory [5, 8], the design patterns for layouts are mostly interpretable via four relations between element-pairs (e.g. Alignment, Repetition, Unity, and Proximity). Therefore, we computationally qualify them based on the design attributes so that each poster can be represented as an **aesthetic-aware graph** (AG) with the novel design relations. Then we conduct k-means [7] on the element-view and relation-view slices on the graphs to obtain several clusters.

3.2 Interactive Poster Design

3.2.1 Diverse Content Expansion. The direct expansion method is to retrieve real-world posters with the input object from each cluster for multiple expansions. However, as users may add more content and re-expand them, ALDA should be capable to deal with complex content collocations, which may not exist in current posters. Therefore we construct an **aggregated knowledge graph** (KG) for each cluster by integrating AGs of corresponding posters to one larger graph, which contains both explicit relations existing in a certain poster and potential relations blended among different posters. Subsequently, we conduct the random walk on each KG, which starts with the user input element and expands one node and the corresponding edge at each step. To adaptively determine when to finish the expanding process, we propose a self-constrained strategy, which calculates a score ϑ_t for the current walking sequence s_t at step t by $\vartheta_t = \sum_{i=1}^N F_{sim}(s_t, g_i)$, where g_i indicates the sequence formulation of the AG of poster i , and F_{sim} calculates the maximum public string ratio for the two sequences indicating their similarity. In other words, the higher the ϑ , the more s_t is consistent with real-world design common sense. When $\vartheta_{t+1} < \vartheta_t$, the walking process is terminated and s_t is reserved as one content expansion. Finally, total content expansions are obtained through the walking process on KGs from all clusters respectively.

3.2.2 Layer-formed Poster Design. To provide users with editable layouts, we propose an aesthetic-aware layout generator (ALG)

Table 1: Evaluations on different layout representation.

Method	Document		Magazine		User Interface	
	V-over $\uparrow$	V-align $\uparrow$	V-over $\uparrow$	V-align $\uparrow$	V-over $\uparrow$	V-align $\uparrow$
AG	0.23	72.86	4.89	65.15	49.11	46.71
SG	0.12	48.43	2.64	28.96	17.46	23.89

based on transformer neural networks [11]. Specifically, we introduce novel design-aware embedding and a design loss function to generate high-quality layouts conditioned on our AGs.

4 EVALUATION

In this section, we test the effectiveness of the proposed ALDA from both quantitative and qualitative evaluations. More experiment results are introduced in the supplementary file.

4.1 Quantitative Evaluation

We conduct k-means on the SGs and AGs respectively to obtain clusters for images, and calculate variances of two common metrics (overlap [4], and alignment [6]) on them, where the higher the variations, the better performances of the clustering. As shown in Table 1, our AGs achieve better performances than SGs.

4.2 Qualitative Evaluation

We invite 20 participants aged from 20 to 50 years ($M = 34.21$, $SD = 6.32$; 45% female) to use our tool, in which 5 participants have formal design education backgrounds and 3 participants are expert designers. The 10-point scale (10 indicates the best) is used to answer the following questions.

Q1: To what extent does ALDA bring convenience to your design process and simply write down the reasons? We got rates from 7 to 10 ($M = 8.91$, $SD = 1.59$) with three main reasons including ‘Provide me with the summarization of patterns’, ‘Can deal with few elements’, and ‘Easy to use’, which prove the effectiveness of our design tools.

Q2: Score for the diversity of expanded contents, and the quality of generated layouts. We got rates from 5 to 9 ($M = 7.36$, $SD = 1.02$) and 6 to 10 ($M = 8.51$, $SD = 0.83$), which indicate 1) AGs proposed in our work validly represent the design pattern information for high-quality clustering and the walking strategy does good for improving the diversity of content expansion and 2) our aesthetic-aware layout generator can output reasonable and appealing results.

5 CONCLUSION

In this work, ALDA aims to assist users with flexible inputs (from a single element to complex collocations) and obtain high-quality posters. Specifically, we propose AGs for effective poster representation and clustering, introduce the self-constrained walking method for more variations, and build a novel layout generator to arrange elements conditioned on AGs.

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